**National University of Computer and Emerging Sciences**

**Lab Manual 07**

Programming Fundamentals

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FAST - School of Computing

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## Activities:

**Note:** Include input validation in programs wherever applicable. Don't use any build-in function.

1. Write a program that helps the user manage their personal budget. The program should prompt the user to input their monthly income and then ask for the total amount they spend on rent, groceries, utilities, transportation, entertainment, and other expenses. Calculate and display the total expenses, the remaining balance, and the percentage of income spent in each category.
2. Write a program that asks the user to enter the current sea level in feet. If the sea level increases by 1.5% each year, your task is to determine the sea level after 2 years, additionally calculate the expected rise in sea level after 2 years.

   Sample Output:

   Enter current sea level: 1000

   Sea level after 2 years: 1030.225

   Increase in the sea level: 30.225
3. Write a program that asks the user to enter the lengths of sides a, b, and c of a triangle. Your task is to calculate angle θ using the following formula:

   $$b^2 = a^2 + c^2 - 2ac\ cos\theta$$

   You can use *cmath* library for acos ($cos^{-1}$) function.
4. Write a program that prompts the user to enter a four-digit positive integer. Make sure that your program should only accept four-digit positive integer. Once the input becomes valid your task is to determine whether the provided integer is a palindrome or not. A palindrome is an integer or a string that remains the same when reversed.
5. Write a C++ program that inputs a positive number from the user and checks whether its the factorial of any number or not. It also asks the user whether user wants to check another factorial or not. If the user inputs 1, it again asks for input and check for another factorial. If the user inputs 0, the program terminates.
6. Write a program that asks for an integer N from the user and prints all prime numbers from 1 till N (both inclusive).

   **Example Input:** 20

   **Example Output:** 2 3 5 7 11 13 17 19
7. Write a program that calculates the harmonic sum of a number N, which is the sum of the reciprocals of the first N natural numbers.

   **Example Input:** 5

   **Example Output:** Sum = 2.28333
8. Write a program that prints the whether the number entered by the user is present in Fibonacci Sequence or not.

   **Example Input:** 13

   **Example Output:** 13 is present in Fibonacci Sequence.
9. Write a program that takes input the number of rows from user and print a hollow square of asterisks (*), where the boundary is filled with * and the inner part is empty:

   **Example Input:** Enter number of rows: 5

   **Example Output:**

```
* * * * *
*       *
*       *
*       *
* * * * *
```

10. Write a program that takes inputs the number of rows from user and prints the number pyramid:

**Example Input:** Enter number of rows: 9

**Example Output:**

```
                1
            1   2   1
        1   2   3   2   1
    1   2   3   4   3   2   1
1   2   3   4   5   4   3   2   1
    1   2   3   4   3   2   1
        1   2   3   2   1
            1   2   1
                1
```